Graphing Quadratics from Vertex or Standard Form

Vertex form, standard form, and transformations of the parent parabola

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- **Vertex form** is $y = a(x-h)^2 + k$: the vertex is (h, k), and the sign of a says which way the parabola opens.
- **Standard form** is $y = ax^2 + bx + c$: the axis of symmetry is $x = -\dfrac{b}{2a}$, the y-intercept is c, and the x-intercepts come from solving $y = 0$.
- Every part is worth writing down: a graph is only as good as the points you found first.
- The last three problems supply a grid. Plot your points, then join them with a *smooth curve* — a parabola never has a sharp corner.

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1. $y = (x-2)^2 - 9$

(a) Find the x-intercepts by solving $(x-2)^2 - 9 = 0$. (b) Find the y-intercept.

Answer: ____________

2. $y = -(x+1)^2 + 4$

(a) Find the x-intercepts. (b) Does this parabola open up or down, and is its vertex a maximum or a minimum? Explain how the equation tells you.

Answer: ____________

3. $y = 2(x-1)^2 - 8$

(a) Find the x-intercepts. (b) Find the y-intercept.

Answer: ____________________

4. $y = x^2 - 6x + 5$

(a) Find the axis of symmetry using $x = -\dfrac{b}{2a}$. (b) Find the y-value of the vertex.

Answer: ____________________

5. $y = x^2 + 4x - 5$

(a) Find the x-intercepts. (b) Find the axis of symmetry.

Answer: ____________________

6. $y = 2x^2 - 8x + 6$

(a) Find the x-intercepts. (b) Find the y-value of the vertex.

Answer: ________________

7. The graph of $y = x^2$ is shifted 3 units to the left and 2 units down.

(a) Write the equation of the new parabola in vertex form. (b) Find the y-intercept of the new parabola.

Answer: ________________

8. The graph of $y = x^2$ is reflected in the x-axis and then shifted 9 units up.

(a) Write the equation of the new parabola. (b) Find its x-intercepts.

Answer: ________________

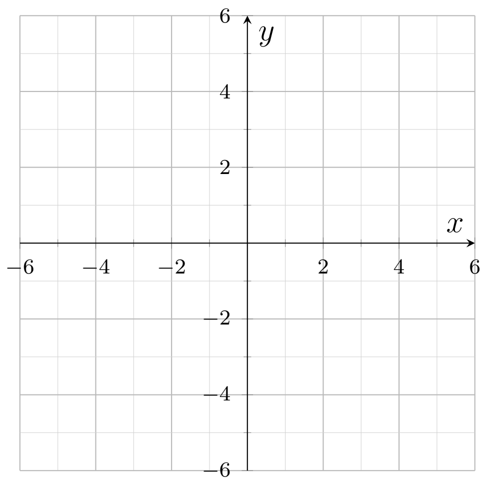
9. $y = -2(x-1)^2 + 8$

(a) Starting from $y = x^2$, describe the three transformations that produce this graph, in the order they are applied. (b) Find the x-intercepts.

Answer: __________________

10. $y = (x-1)^2 - 4$

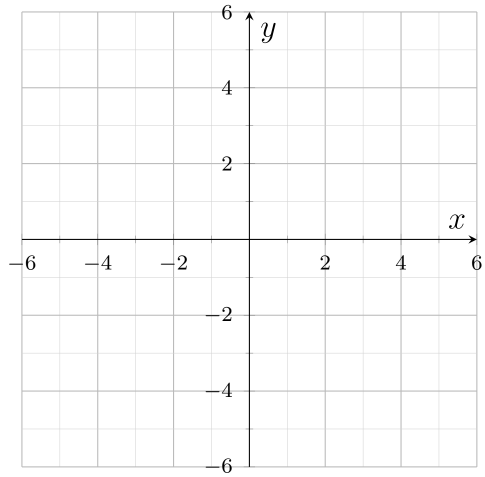
(a) Find the x-intercepts. (b) Plot the vertex and both x-intercepts on the grid, then sketch the parabola.



Answer: __________________

11. $y = -x^2 + 2x + 3$

(a) Find the x-intercepts. (b) Find the y-intercept. (c) Plot those points and the vertex, then sketch the parabola.

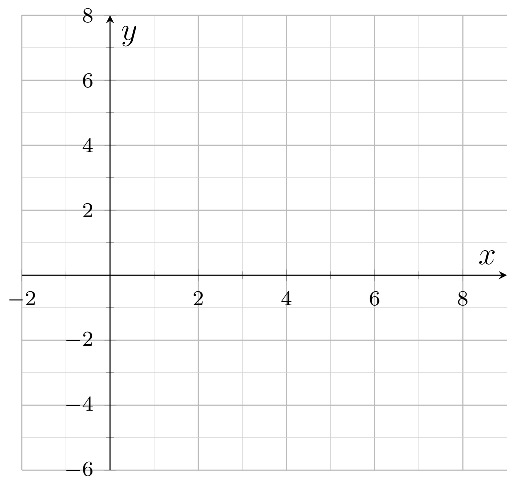


Answer: ____________________

12. Synthesis challenge. $y = x^2 - 8x + 12$

(a) Find the axis of symmetry. (b) Find the y-value of the vertex. (c) Find the x-intercepts.

(d) Sketch the parabola on the grid, then explain how the axis of symmetry lets you check that your two x-intercepts are correct.



Answer: ____________________